

PHASE 6

PROJECT TESTING PHASE

Testing Objectives

- To verify that the narrative generation utilizing Gemini models (`gemini_pro.py` and `gemini_flash.py`) functions correctly based on user input.
- To ensure the `image_generator.py` successfully synthesizes corresponding panel artwork using local diffusion models.
- To confirm that the `layout_builder.py` correctly assembles text and images into cohesive, structured comic pages.
- To validate that the `exporters.py` module successfully formats and saves the final layout as a PDF document into the `static/exports/` directory.
- To ensure secure loading of API keys and configurations via the `.env` file and `config.py`.

Testing Environment

- **Backend Framework:** Python environment configured with dependencies from `requirements.txt` and `requirements-local-diffusion.txt`.
- **Frontend:** Web browsers interacting with the HTML/CSS interface via `main.py` and `routes.py`.
- **Security/Config:** Local environment containing the `.env` file for secure API key and path management.

Testing Methods

- **Unit Testing:** Isolating and validating individual modules, specifically the text generator, image generator, and layout builder.
- **Integration Testing:** Verifying the data flow between the Gemini API (text/prompts) and the local diffusion environment (images) into the final layout structurer.
- **System/End-to-End Testing:** Running the complete pipeline from web interface prompt submission to final PDF download.

Final Testing Result The ComicCraft application is stable, secure, and functionally complete according to the initial project requirements. The system is fully prepared for the Phase 8 demonstration.